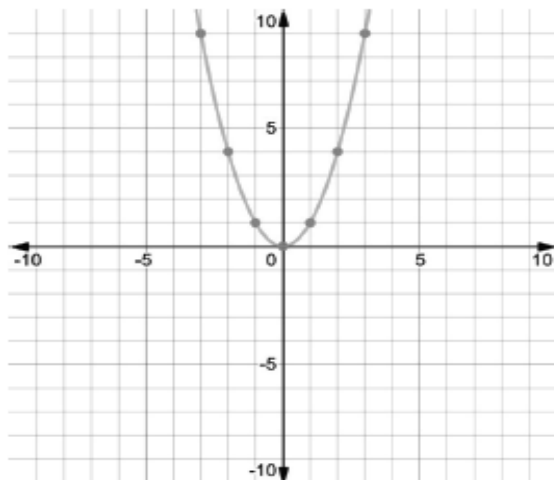


Algebra 1
Unit 1
Lesson 1 Practice

Name _____
Date _____
Period _____

1. A graph is shown.



The graph is the parent function for a(n)

- ☐ linear function.
- ☐ quadratic function.
- ☐ cubic function.
- ☐ square root function.
- ☐ cube root function.
- ☐ absolute value function.
- ☐ exponential function.

2. The equation $y = \sqrt{x}$ is the parent function for a(n)

- ☐ linear function.
- ☐ quadratic function.
- ☐ cubic function.
- ☐ square root function.
- ☐ cube root function.
- ☐ absolute value function.
- ☐ exponential function.

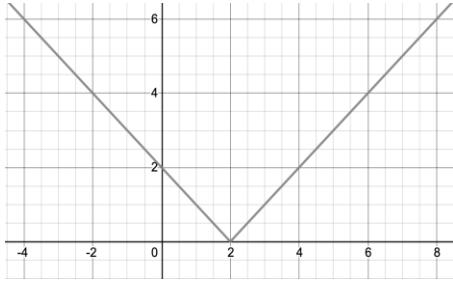
3. The equation $y = |x|$ is the parent function for a(n)

- ☐ linear function.
- ☐ quadratic function.
- ☐ cubic function.
- ☐ square root function.
- ☐ cube root function.
- ☐ absolute value function.
- ☐ exponential function.

For equations 4-10, classify the function family for each equation or graph.

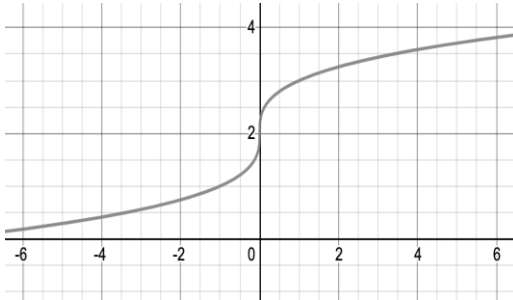
4. $y = \frac{3}{5}x - 9$

5.

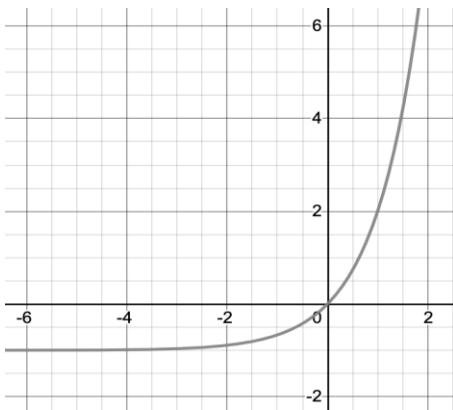


6. $y = -3\left(\frac{1}{2}\right)^x$

7.



8.



9. $y = (x - 3)^3 + 2$

10. $y = -4\sqrt{x} + 1$

Algebra 1
Unit 1
Lesson 2 Practice

Name _____
Date _____
Period _____

Given the function $r(x) = 5x - 3.5$, find each value.

1. $r(3) =$

2. $r\left(6\frac{1}{2}\right) =$

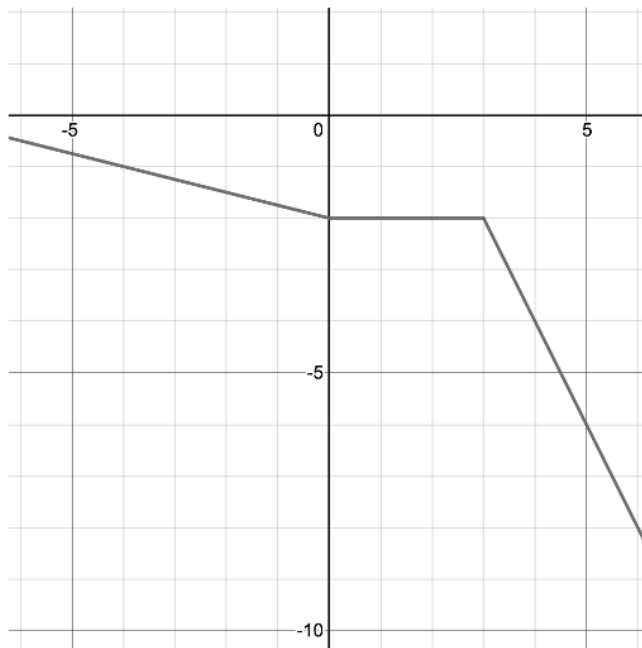
3. For what value of x is $r(x) = -21$?

Use the graph of $y = f(x)$ to find each value.

4. $f(-4) =$

5. $f(1) =$

6. For what value of x is $f(x) = -4$?



Use the table of values representing $m(x)$ to find each value.

7. $m(-4) =$

x	$m(x)$
7	2
-4	-9
0	8
-15	9
4	-5
6.5	0

8. $m(4) =$

9. What is the y -intercept of the function in the table?

10. Given $d(x) = 3x^3 - 4x^2 + 2$, evaluate $d(x)$ for $x = -1$.

Algebra 1
Unit 1
Lesson 3 Practice

Name _____
Date _____
Period _____

The function $G(m) = 49.5m + 89.75$ represents the cost of joining the gym in addition to the one time membership fee. The cost, G , is measured in dollars for m months.

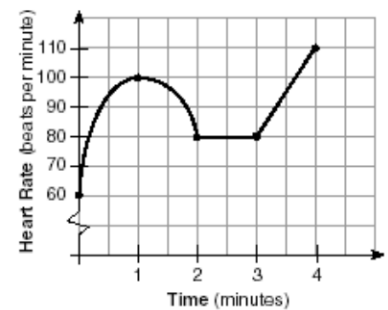
1. Find the value of $G(6)$.
2. What does $G(6)$ represent in the context of the problem?
3. Is the value of $G(9.75)$ reasonable to interpret in the context of the problem?

A function $H(m)$ is modeled in the graph below. It shows the heart rate of a swimmer during a 4 minute race where m is the number of minutes and H is the heart rate of the swimmer.

4. Find the approximate value of the graph for $H(1.5)$.

$$H(1.5) \approx$$

5. What does $H(1.5)$ mean in the context of the problem?



6. After how many seconds is the swimmers heart rate 80 beats per minute?
7. Why does it make sense that $H(2) > H(0)$ in this context?

The table below represents the temperature of a cup of coffee as it begins to cool. Temperature readings are taken every 5 minutes and measured in degrees Fahrenheit ($^{\circ}\text{F}$).

Minutes, m	0	5	10	15
Temperature, $T(m)$	197	186	170	164

8. Explain the meaning of $T(0) = 197$.
9. Determine the value of $T(10)$.
10. Interpret $T(10)$ for the given context.

Algebra 1
Unit 1
Lesson 4 Practice

Name _____
Date _____
Period _____

For questions 1-5, use the exponent laws to write equivalent expressions.

1. $\left(\frac{6}{5}\right)^{-2} =$

2. $\frac{7^7}{7^4} =$

3. $(5.2 \cdot (-9))^3 =$

4. $(-2) \cdot (-2)^{-5} \cdot (-2)^8 =$

5. $((-4)^4)^9 =$

Find the value of each for questions 6-7.

6. $\sqrt[3]{125} =$

7. $\sqrt{25} =$

8. Explain why both the $\sqrt{25}$ and $\sqrt[3]{125}$ both equal 5.

For questions 9-10, find the value of the variable that makes the equation true.

9. $\sqrt{c} \cdot \sqrt{c} = 11$

$c = \underline{\hspace{2cm}}$

10. $\frac{2^3}{3^3} = \left(\frac{3}{2}\right)^g$

$g = \underline{\hspace{2cm}}$

Algebra 1
Unit 1
Lesson 5 Practice

Name _____
Date _____
Period _____

1. In radical form, the expression $x^{\frac{1}{3}}$ is equivalent to _____.
2. In exponential form, the expression $\sqrt{x}$ is equivalent to _____.

For questions 3-5, rewrite each exponential expression as a radical expression. Then, evaluate the expression.

3. $(-729)^{\frac{1}{3}} =$

4. $(243)^{\frac{1}{5}} =$

5. $(225)^{\frac{1}{2}} =$

6. Simplify the expression using the laws of exponents and then find the value.

$$(-27)^{-\frac{2}{9}} \cdot (-27)^{\frac{5}{9}} =$$

The expression $\frac{(-125)^{\frac{11}{9}}}{(-125)^{\frac{8}{9}}}$ can be rewritten using the law of exponents.

7. Which law of exponents can be used to rewrite the expression?

8. Rewrite the expression as an equivalent expression.

9. Write the simplified expression in radical form.

10. Evaluate the radical expression.

Algebra 1
Unit 1
Lesson 6 Practice

Name _____
Date _____
Period _____

1. Explain mathematically why the expressions $\left(a^{\frac{1}{c}}\right)^b$ is equivalent to $(a^b)^{\frac{1}{c}}$ using the expressions $\left(4^{\frac{1}{2}}\right)^5$ and $(4^5)^{\frac{1}{2}}$.

2. Identify all of the expressions equivalent to $\left(49^{\frac{1}{2}}\right)^5$.

- $49^{\frac{2}{5}}$
- $(49^5)^{\frac{1}{2}}$
- $\sqrt[5]{49^2}$
- $49^{\frac{5}{2}}$
- $(\sqrt{49})^5$

For questions 3-7, complete each row with equivalent expressions.

	$a^{\frac{b}{c}}$	$\left(a^{\frac{1}{c}}\right)^b$	$(a^b)^{\frac{1}{c}}$	$(\sqrt[c]{a})^b$	$\sqrt[c]{a^b}$
3.		$\left(256^{\frac{1}{4}}\right)^3$			
4.					$\sqrt{64^3}$
5.	$128^{\frac{5}{7}}$				
6.			$(-27^2)^{\frac{1}{3}}$		
7.				$(\sqrt{81})^3$	

For questions 8-10, rewrite each expression using a rational exponent in the form $(a)^{\frac{b}{c}}$.

8. $\sqrt[3]{(-44)^5}$

9. $(\sqrt{289})^3$

10. $\frac{1}{\sqrt[5]{30^2}}$

Algebra 1
Unit 1
Lesson 7 Practice

Name _____
Date _____
Period _____

1. Write the expression $27^{\frac{2}{3}}$ in two different equivalent forms.

Expression 1	Expression 2

2. The value of $27^{\frac{2}{3}}$ is

- ☐ 9
- ☐ 32
- ☐ 54
- ☐ 72

For questions 3-5, write an equivalent radical expression and then find the value of the expression.

3. $(-125)^{\frac{2}{3}} = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$

4. $-\left(\frac{3,125}{32}\right)^{\frac{2}{5}} = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$

5. $(81)^{-\frac{3}{2}} = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$

Evaluate each expression in questions 6-10 without a calculator.

6. $(4^{-2})(49^{-\frac{1}{2}})$

7. $(256^{\frac{1}{4}})^3 \cdot 6^{-1}$

8. $-\left(\frac{1}{4}\right)^{-\frac{7}{2}}$

9. $(-1^{15}) \cdot (-64^{\frac{1}{3}})$

10. $\left(\frac{5}{35}\right)^{-2} \cdot (2^{-2})$

Algebra 1
Unit 1
Lesson 8 Practice

Name _____
Date _____
Period _____

1. Which of the following is not a radical expression that is equivalent to $\sqrt{96}$?
 - A. $\sqrt{2} \cdot \sqrt{48}$
 - B. $\sqrt{5} \cdot \sqrt{19}$
 - C. $\sqrt{6} \cdot \sqrt{16}$
 - D. $\sqrt{8} \cdot \sqrt{12}$

2. Which of the expressions from #1 can be rewritten in the form of an integer multiplied by the radical? *Explain.*

For questions 3-4, write two equivalent expressions for each radical expression using the product law of radicals. One of your expressions should simplify to an integer multiplied by a radical.

3. $\sqrt[3]{192} =$ _____

$\sqrt[3]{192} =$ _____

4. $\sqrt{80} =$ _____

$\sqrt{80} =$ _____

5. Rewrite $\sqrt[3]{192}$ as an integer multiplied by a radical.

6. Rewrite $\sqrt{80}$ as an integer multiplied by a radical.

7. Identify all expressions that are equivalent to $2\sqrt[3]{54}$.

- $6\sqrt[3]{2}$
- $23\sqrt[3]{2}$
- $6\sqrt[3]{6}$
- $2 \cdot \sqrt[3]{2} \cdot \sqrt[3]{27}$
- $2 \cdot \sqrt[3]{6} \cdot \sqrt[3]{9}$

For questions 8-10, write an equivalent expression in the form of a constant multiplied by a radical.

8. $2\sqrt{200}$

9. $\sqrt[3]{40}$

10. $8\sqrt[3]{(-500)}$

Algebra 1
Unit 1
Lesson 9 Practice

Name _____
Date _____
Period _____

1. Which of the following expressions can be combined?

- A. $2\sqrt{80} - 6\sqrt{20}$
- B. $3\sqrt{6} + 2\sqrt{12}$
- C. $5\sqrt{18} - 18\sqrt{5}$
- D. $-4\sqrt{50} + 5\sqrt{40}$

Three students simplified the expression $2\sqrt{90} - 7\sqrt{40}$ below. Their work is shown below.

Amy	Linda	Jaxson
$2\sqrt{90} - 7\sqrt{40}$ $2\sqrt{(9 \cdot 10)} - 7\sqrt{(4 \cdot 10)}$ $6\sqrt{10} - 14\sqrt{10}$ $-8\sqrt{10}$	$2\sqrt{90} - 7\sqrt{40}$ $-5\sqrt{50}$	$2\sqrt{90} - 7\sqrt{40}$ $2\sqrt{(9 \cdot 10)} - 7\sqrt{(4 \cdot 10)}$ $18\sqrt{10} - 28\sqrt{10}$ $-10\sqrt{10}$

2. Which student simplified the expression correctly?

3. Write a note to one of the students who made an error explaining their mistake.

4. Write a note to the other student who made an error explaining their mistake.

For question 5-10, find the sum or difference of the expression. If you cannot add or subtract the expression, explain why.

5. $\sqrt{18} - 4\sqrt{2}$

6. $\sqrt[3]{128} - \sqrt[3]{54}$

7. $7\sqrt{47} + 8\sqrt{147}$

8. $125^{\frac{1}{2}} + 320^{\frac{1}{3}}$

9. $\sqrt{200} - 2\sqrt{48} + 4\sqrt{98}$

10. $192^{\frac{1}{3}} + 2\left(81^{\frac{1}{3}}\right)$

Algebra 1
Unit 1
Lesson 10 Practice

Name _____
Date _____
Period _____

1. To multiply two _____ with the same index, multiply the integers together and then multiply the radicands together. Then simplify the radical expression.

For questions 2-7, simplify each radical expression by writing an equivalent radical expression.

2. $\sqrt{11} \cdot \sqrt{11}$

3. $\frac{3\sqrt{3}}{-\sqrt{48}}$

4. $\sqrt[3]{8} \cdot \sqrt[3]{16}$

5. $\sqrt{10} \cdot \sqrt{6}$

6. $\frac{\sqrt[3]{432}}{\sqrt[3]{128}}$

7. $\sqrt[3]{98} \cdot \sqrt[3]{21}$

8. Identify all equivalent expressions to $\sqrt{7} \cdot \sqrt{14}$.

- $\sqrt{98}$
- $7\sqrt{7}$
- $(7 \cdot 14)^{\frac{1}{2}}$
- $7\sqrt{2}$
- $14\sqrt{2}$
- $7^{\frac{1}{2}} \cdot 14^{\frac{1}{2}}$

For questions 9-10, write an equivalent expression by rationalizing the denominator.

9. $\frac{-5\sqrt{3}}{2\sqrt{5}}$

10. $\frac{\sqrt{27}}{3\sqrt{6}}$